

Final Report: FPGA-Based Fighting Game : FOOTSIES

MEHMET BAŞTÜRK 2574762— EMİN ÖZBERK AKKAN 2574473— YUSUF BERAT BÖLÜKBAŞ 2574804

Department of Electrical and Electronics Engineering
Middle East Technical University
Ankara, Turkey

Abstract—This report describes the design, implementation, and implementation of the term project, two-player fighting game by using the DE1-SoC FPGA board inputs. This game is a real-time system created using Verilog HDL through Intel Quartus. This project is inspired by an open-source fighting game called "Footsies" which is based on the FPGA-based FPGA. The game includes basic character movement and attacks to more complex mechanics like hitbox-based collision detection, blockstun, hitstun and a CPU opponent. The project provides invaluable hands-on experience in digital system design and the integration of multiple hardware modules into a cohesive, real time interactive system.

Index Terms—FPGA, Verilog, Digital Design, Fighting Game, VGA, State Machine, Hitbox, Hurtbox.

I. INTRODUCTION

The objective of this project was to code, implement and create a two player footsies game by using hardware language Verilog HDL. The game is played by using the inputs of the FPGA and external membran keypad set through the Video Graphics Array (VGA) output. The project's goal was to integrate hardware mechanics to a visual output and make students learn the Verilog HDL and hardware applications by this entertaining project. Also it helps to comprehend the integration of complex state machines into the programming languages.

The project was inspired by an open-source fighting game called "Footsies". The game has two game modes, playing against another player or bot. Each character has actions by the inputs such as moving left, right, attacking, and blocking. The objective is hit the opponent until it has zero health in 100 seconds. After (at least) one of the characters have zero health, game is over and the result will be displayed on the screen.

II. THEORETICAL BACKGROUND

This project leverages several key digital design concepts:

- **Finite State Machines (FSMs):** The game's logic depends on a complex state machine for each player state such that MoveLeft, MoveRight, Attack etc. These states are visualised on the screen by 60Hz VGA connection. This state machine also connected with some rules which includes some input combinations of the player(s).
- **Input Handler and Debouncer:** Players give inputs to the FPGA through the external devices (buttons and keypads). These inputs are combined with the FSM modules,

and form the game logic. Also includes a debouncer for the button inputs which adds a small delay to the inputs to ensure the input accuracy.

- **Memory:** For the external visuals such as MENU, Health / Block bar and the end game output, we utilized Read Only Memory (ROM).
- **Hit/Hurtbox:** Most of the game rules depends on this visual hit/hurtboxes. Depending on the state of the character, players can lose health or block point.
- **Clock Division:** FPGA's 50 MHz clock was divided to 25 MHz and 60Hz to supply VGA and the game logic.

III. PROPOSED SOLUTION

The system has Verilog modules which process player inputs with game logic for VGA output.

A. System Architecture

A top level module instantiates following modules:

- **Game Logic Module:** An FSM that tracks the state of each player, health and block points, and game state (menu, gameplay, game over).
- **Player State Module:** Two separate modules to handle debounced inputs from each player's keypad for moving left, right, and attacking.
- **VGA Controller and Driver Module:** Manages the generation of VGA sync signals and pixel data. A pre-existing, properly referenced controller was utilized for this purpose.
- **Clock Divider Modules:** Provides required clock variables which are 25 MHz and 60 Hz for both VGA and game logic modules.
- **Rendering Module:** Corresponding to visual output module for the project including menu, health bar player sprites and game result displays.
- **7-Segment Display and LED Drivers:** Managing LEDs on the FPGA and Displays to inform the gamers for the current state of both game and the player(s).

B. Project Phases

The game has three phases: Menu, Gameplay, and Game Over.

- **Menu Phase:**
 - The game starts in this state.

- Seven-segment display shows "1P" or "2P".
- Player 1 presses a button to start the selected game mode and game goes into the "Gameplay" phase .

- **Gameplay Phase:**

- Gameplay Phase starts with "3, 2, 1, START" count-down on the screen.
- Seven-segment displays "FIGHT" when the game starts.
- FSM loop starts and the states are upgraded until the next phase for corresponding inputs.
- The game continues until one of the characters at 0 Health at the screen.
- When the game is over, game goes into the "Game Over" phase.

- **Game Over Phase:**

- The screen displays the winner and the time stops.
- All LEDs blink.
- Seven-segment displays the result and match duration.
- Player 1 returns to the Menu Phase again.

C. Character States

The FSM of the characters has the following states:

- **IDLE:** The default state of the characters.
- **MOVING FORWARD/BACKWARD:** The character moves at 3 pixel per frame FORWARD or 2 pixel per frame BACKWARD.
- **ATTACK STARTUP:** Attack started. At the next state character will be attacking.
- **ATTACK ACTIVE:** Character is attacking. There is an hurtbox which can damage the other character.
- **ATTACK RECOVERY:** Character attacked and will wait for the recovery.
- **HITSTUN:** Brief stun state after getting hit.
- **BLOCKSTUN:** Brief stun state after blocking an attack.

IV. RESULTS

The project is created and implemented on the DE1-SoC board. It has successfully achieved all the given conditions for the project.

Player FSM : Player State Machine is fully functional as in the testbench result , figure 1.

Game Logic : Project is completely successful on meeting the game rules. When the hurtbox touch the hitbox, player gets damage. Also blocking is a working mechanism.

Visuals : Menu is working perfectly. It can be easily observed the game mode differences. Health / Block Bar fully connected with the logic. Also counter and end game information works properly. In addition, LED displays are fully compatible with the logic. As you can see in Fig. 2, We have a clear menu phase. Also in figure 3 , we have an example winner end game phase. Also in figure 4, you can see a picture of the gameplay phase. In figure 5, LED Display shows the "Fight" word to inform players are in game.

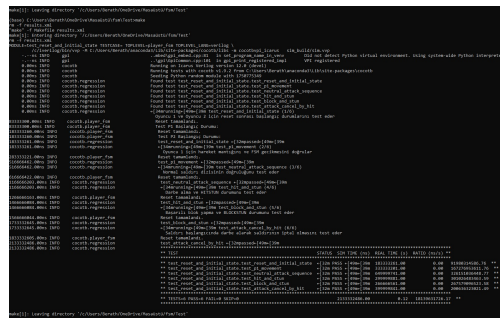


Fig. 1. Player FSM Testbench Result

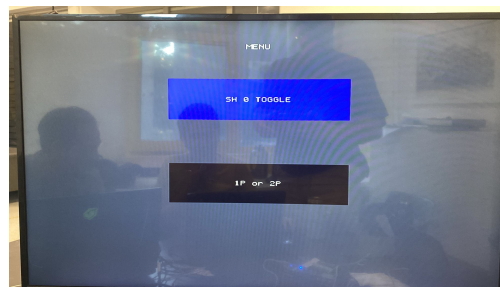


Fig. 2. MENU Phase

Inputs : Input of the system works properly, both in terms of hardware and software. Also keypad is clearly functional with the player 2 in PvP mode.

Frames : Game has corrected frame rates in all states of the characters. Also frame counting mode works properly which is helpful to count all frames in the game.

Overall, project is completely functional.

V. CHALLENGES AND DISCUSSION

Some technical challenges or problems were faced during the project both in terms of software and hardware.

- **Game Logic - Input Integration:** Game logic is hard to make it open for every input combination. Some delay problems were faced and solved by the input debouncer. Disabling reset button in frame counting mode can be exemplified as a problem.
- **State Machine Complexity:** Player FSM became complex due to the number of character states and transitions. Directing some states and displays together caused some

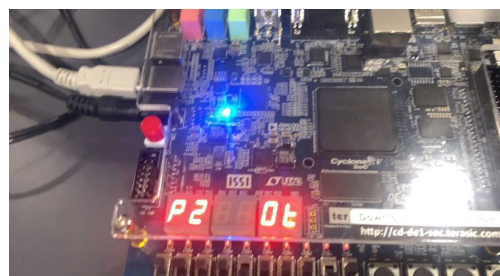


Fig. 3. P2 Wins LED Display

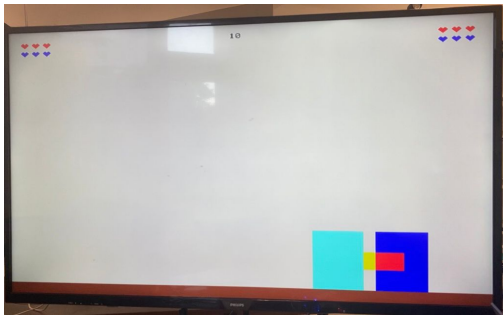


Fig. 4. Gameplay Phase Interaction Example

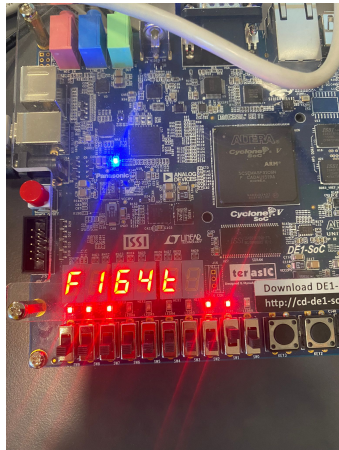


Fig. 5. LED Display in Gameplay Phase

The project is an important milestone for combining different design lecture topics both in digital and logic. Moreover, this project experienced more than design. It has a considerable hardware-software intersection that makes us more proficient on electronics.

REFERENCES

- [1] HiFight, "FOOTSIES," 2018. Accessed: 2025-05-10. [Online]. Available: [1] in project description
- [2] V. H. Adams, "Vga driver in verilog," 2025. Accessed: 2025-05-10. [Online]. Available: [2] in project description

problems in terms of number of inputs and outputs for top module.

- **Hit Detection Logic:** Managing VGA Graphics with the game logic requires hit detection which was a little bit problematic for the Player FSM integration. For example ; after blocking the attack , player should be in BLOCK-STUN state in the Player FSM.
- **Input Handling:** Input was a little bit problematic at the start. Placing a debouncer solved the problem.
- **Attack Direction :** At the beginning , attack direction could be changed by the move inputs. Then we removed the selecting direction mechanic. After removing this mechanic, system got some errors in Player FSM.
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These challenges were handled by using extended problem solving methods all through the project. These are the most common problems that are encountered. Note that all through the project we faced hundreds of problems even the basic syntax errors in our codes.

VI. CONCLUSION

This project reached the term project goal which is 2 player fighting game with FPGA and Verilog HDL . The final design of the project has specific mechanics and features like attack , directional attack or block against the CPU bot or enemy player, depends on your mode selection in menu.